

SEBA Master: Web Application Engineering

1. Web Site Genres

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1. Web Site Genres

- Business Models on the Web
- Business Model Canvas & Value Proposition Canvas

The material for this unit has been taken from the course "Managing the Digital Enterprise" of Prof. Michael Rappa from North Carolina State University.

[Ra11] Rappa, Michael, "Business Models on the Web", <https://digitalenterprise.org/models/>, [Last Accessed: April 14th, 2023]

- A business model is the method of doing business by which a company can sustain itself
 - that is, **generate revenue**.
- The business model spells out how a company makes money by specifying where it is positioned in the **value chain**.

Simple example:

- A company produces a good and sells it to customers.

More complicated:

- Radio and Television Broadcasting
 - The broadcaster is part of a complex network of distributors, content creators, advertisers (and their agencies), and listeners or viewers.
 - Who makes money and how much is not always clear at the outset.

source: [Ra11]

Why are business models on the web important / different?

- In general, there is a limited, well understood set of models businesses can make money with.
- The web didn't invent entirely new ways to make money but tried-and-true models have been reinvented, e.g., auctions.
- The web has popularized the auction model and broadened its applicability to a wide array of goods and services.

Especially on the web, business models must be considered carefully:

- During the dot-com bubble most web companies did not have a sustainable business model.

source: [Ra11]

Basic Categories of Business Models on the Web

- **Brokerage**
 - **Merchant**
 - **Manufacturer (Direct)**
 - **Advertising**
 - **Subscription**
 - **Utility**
 - **Community**
 - **Affiliate**
 - **Infomediary**
-
- The models are implemented in a variety of ways, as described later with examples.
 - A firm may combine several different models as part of its overall Internet business strategy.

source: [Ra11]

- Brokers are **market-makers**: they bring buyers and sellers together and facilitate transactions.
- Brokers play a frequent role in:
 - business-to-business (B2B),
 - business-to-consumer (B2C),
 - consumer-to-consumer (C2C) markets.
- Usually, a broker charges a **fee or commission for each transaction** it enables.
- **The formula for fees can vary:**
 - Who pays the fee (the buyer, or the seller, or both?)
 - How much is the fee (a percentage of the transaction volume, a flat fee?)
- The broker is **acting as a third party in the process of the transaction**.

source: [Ra11]

Variations of the Brokerage Model

Auction Broker (e.g., eBay)

- Conducts auctions for sellers (individuals or merchants).
- Broker charges the seller a listing fee and commission scaled with the value of the transaction.
- Auctions vary widely in terms of the offering and bidding rules.

Transaction Broker (e.g., PayPal)

- Provides a third-party payment mechanism for buyers and sellers to settle a transaction.

Demand Collection System (e.g., priceline.com)

- The patented "name-your-price" model pioneered by Priceline.com.
- Prospective buyer makes a final (binding) bid for a specified good or service, and the broker arranges fulfillment.

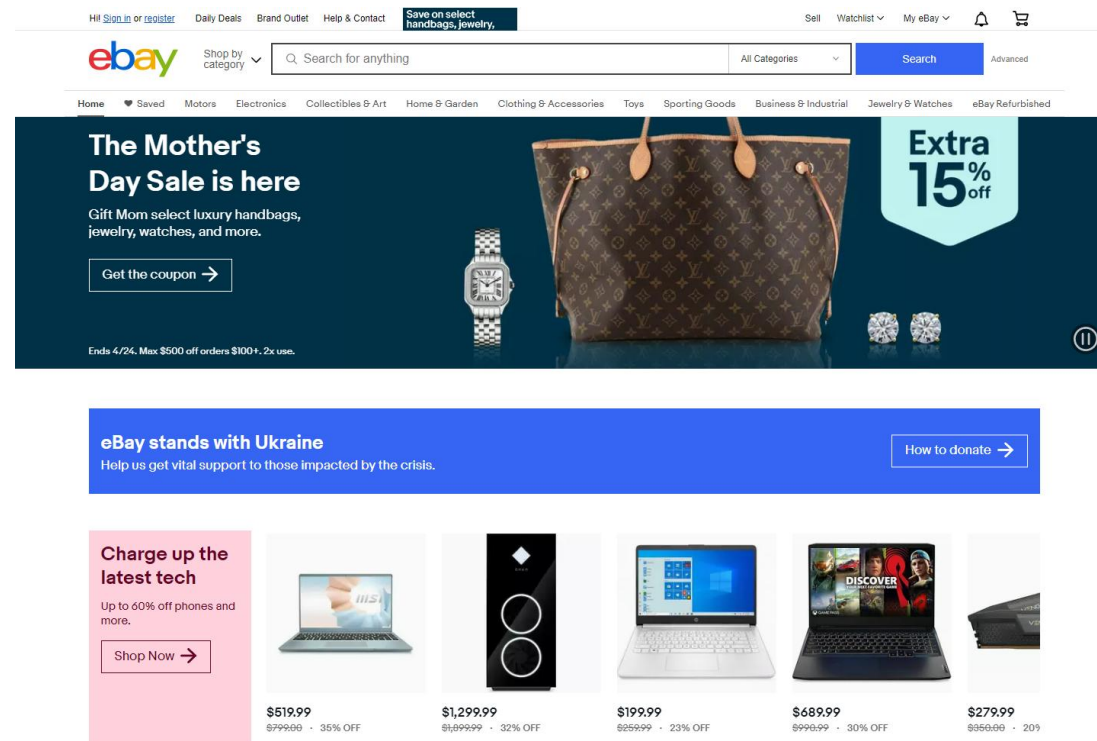
Mobility Broker (e.g., Share Now)

- Assembles all mobility offers of a certain region.
- Provides mobility services like car-sharing, bike-sharing or intermodal traveling, etc.

source: [Ra11]

Case Study: eBay

- Started in 1995, highly profitable from the start.
- In 2004: 1,4 billion listings of products, 168 million registered users, 68 million active users.
- *Challenge*: limited ability to guarantee satisfaction, since eBay only acts as a broker.
- Actions to reduce fraud:
 - Focus on the "eBay community" with certain standards and values.
 - Feedback mechanism: rating for each transaction from both the seller's and the buyer's perspective.



source: [Ra11]

Merchant Model

- Wholesalers and retailers of goods and services.
- Sales may be made based on list prices or through auction.
- A merchant usually creates an **inventory**.
- A merchant is involved in **distribution**.

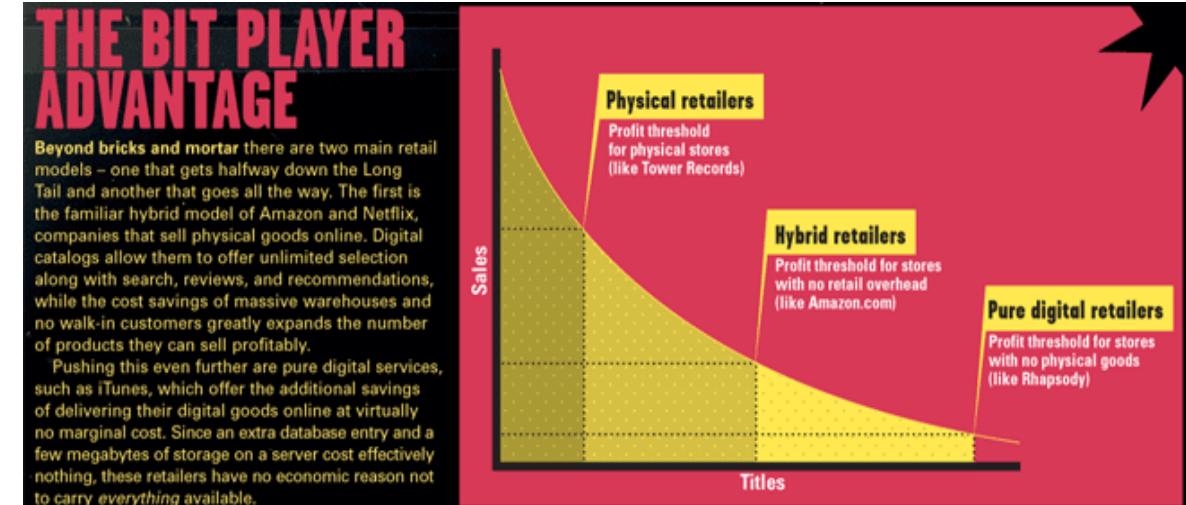
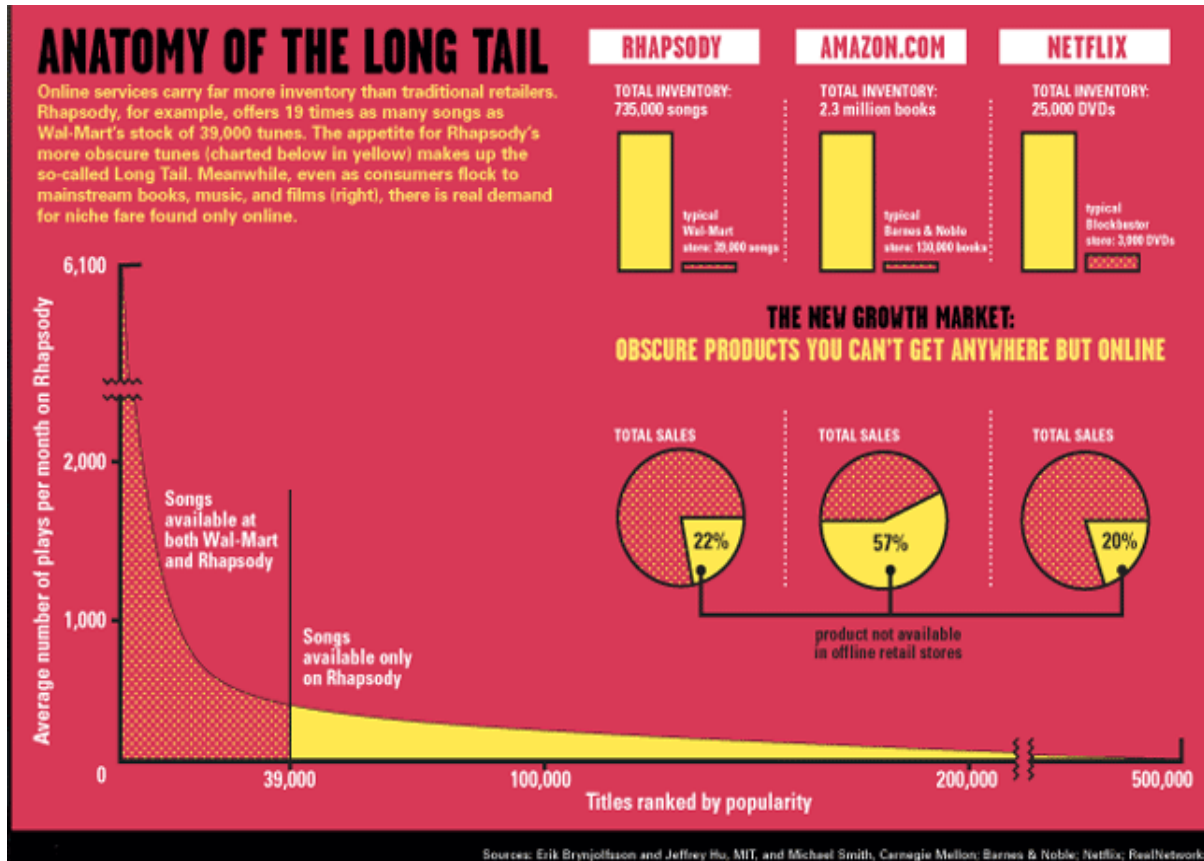
source: [Ra11]

Variations of the Merchant Model

- **Virtual Merchant** (e.g., Amazon)
 - or e-tailer, is a retail merchant that operates solely over the web.
- **Bit Vendor** (e.g., Apple iTunes Music Store, Google Play Store, Steam)
 - A merchant that deals strictly in digital products and services and, in its purest form, conducts both sales and distribution over the web.
- **Catalog Merchant** (e.g., Otto)
 - Mail-order business with a web-based catalog. Combines mail, telephone and online ordering.
- **Click and Mortar** (e.g., Barnes & Noble)
 - Traditional brick-and-mortar retail establishment with web storefront.

The Long Tail

- Online services carry far more inventory than traditional retailers.
- Just as lower prices can entice consumers down the Long Tail, recommendation engines drive them to obscure content they might not find otherwise.



["The Long Tail: Why the Future of Business Is Selling Less of More", Chris Anderson, Hachette Books, 2008]

- The manufacturer or "direct model" is predicated on the power of the web to allow a manufacturer (i.e., a company that creates a product or service) to **reach buyers directly and thereby compress the distribution channel**.
- The manufacturer model can be **based on efficiency, improved customer service, and a better understanding of customer preferences**.
- Prominent Examples: Dell, Spreadshirt
- Difficulty: Co-existence with indirect distribution channels (Media Markt, Saturn, ...)

source: [Ra11]

Variations of the Manufacturer (Direct) Model

Purchase (e.g., Dell)

- The sale of a product in which the right of ownership is transferred to the buyer.

Lease (e.g., Sixt)

- In exchange for a rental fee, the buyer receives the right to use the product under a "terms of use" agreement.
- The product is returned to the seller upon expiration or default of the lease agreement. One type of agreement may include a right of purchase upon expiration of the lease.

License (e.g., CNET Download.com)

- The sale of a product that involves only the transfer of usage rights to the buyer, in accordance with a "terms of use" agreement.
- Ownership rights remain with the manufacturer (e.g., with software licensing).

source: [Ra11]

Add to Cart

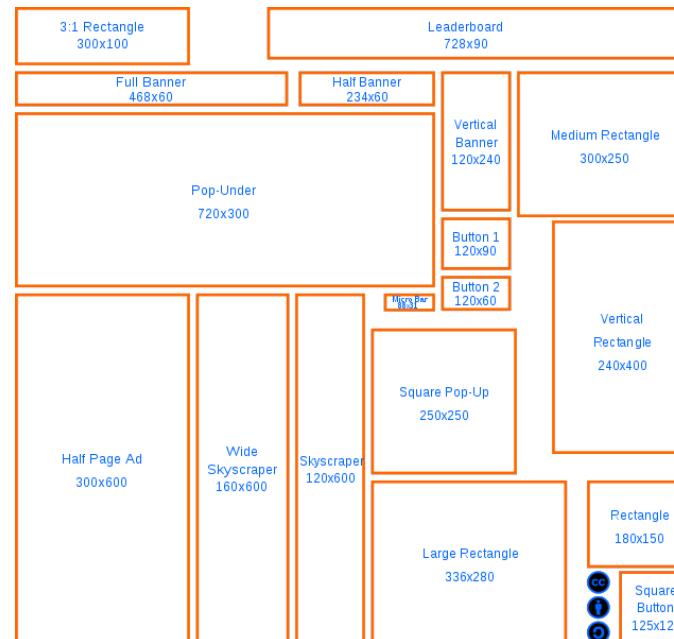


16GB 4267MHz
LPDDR4x Memory
Onboard

Advertising Model

- Is an extension of the traditional media broadcast model.
- The broadcaster, in this case, a web site, provides content (usually, but not necessarily, for free) and services (like email, IM, blogs) mixed with advertising messages in the form of banner ads.
- In the web **the effect of advertisement can be analyzed immediately**.
- The banner ads may be the major or sole source of revenue for the broadcaster.
- The broadcaster may be a content creator or a distributor of content created elsewhere.
- The advertising model **works best when the volume of viewer traffic is large or highly specialized**.

Standardized Sizes of Banner Ads



source: [Ra11]

Variations of the Advertising Model (1)

Portal (e.g., Yahoo)

- Usually, a search engine that may include varied content or services.
- A high volume of user traffic makes advertising profitable and permits further diversification of site services.
- A personalized portal allows customization of the interface and content to the user.
- A niche portal cultivates a well-defined user demographic.

Query-based Paid Placement (e.g., Google)

- Sells favorable link positioning (i.e., sponsored links) or advertising keyed to particular search terms in a user query, such as Overture's trademark "pay-for-performance" model or Google's "Google Ads".

Location-based Placement (e.g., TripAdvisor)

- Provide location-specific advertisements mainly on mobile devices.
- Push location-based service: User receives promotional materials from advertisers nearby.
- Pull location-based service: User searches for places of interests nearby, like restaurants, sightseeing, etc.

source: [Ra11]

Variations of the Advertising Model (2)

Classifieds (e.g., Craigslist)

- List items for sale or wanted for purchase.
- Listing fees are common, but there also may be a membership fee.

Content Targeting (e.g., Google AdSense)

- Pioneered by Google, it extends the precision of search advertising to the rest of the web.
- Google identifies the meaning of a web page and then automatically delivers relevant ads when a user visits that page.

source: [Ra11]

Subscription Model

- Users are charged a **periodic** (daily, monthly, annual, etc.) fee to subscribe to a service.
- Subscription fees are incurred **irrespective of actual usage rates**.
- Subscription and advertising models are frequently combined.

source: [Ra11]

Variations of the Subscription Model

Content Services (e.g., Netflix, Spotify)

- Provide text, audio, or video content to users who subscribe for a fee to gain access to the service.
- On the web it is possible to create a prolific relationship with the customers.

Person-to-Person Networking Services (e.g., Classmates)

- Are conduits for the distribution of user-submitted information, such as individuals searching for former schoolmates.

Internet Services Providers (e.g., Verizon)

- Offer network connectivity and related services on a monthly subscription.

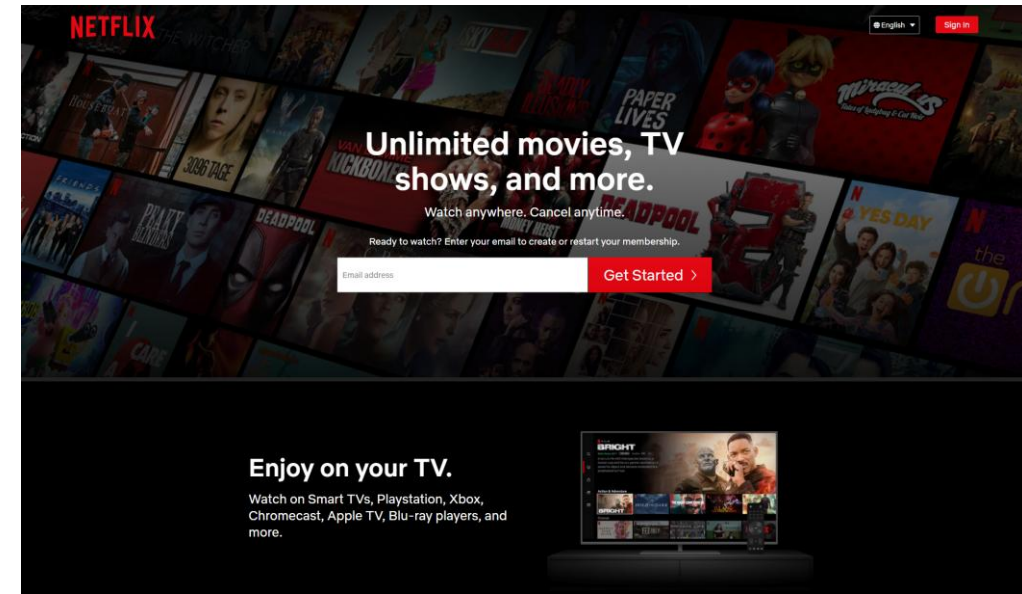
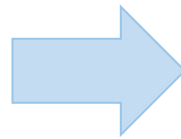
Trust Services (e.g., TrustArc)

- Come in the form of membership associations that abide by an explicit code of conduct, and in which members pay a subscription fee.

source: [Ra11]

Case Study: Netflix.com

- Founded in 1997: Reed Hastings, the founder of Netflix had to pay a late fee for a rented movie.
- Motivation: **Late fees annoy customers** but are necessary to make the classical movie rental system work.
- In the past: Club of movie viewers would pay a membership fee to be allowed to borrow a certain number of DVDs (3) for **as long as they wished**; when returned they would receive the **next movie on the queue**.
- Now: Subscription fee for unlimited online streaming access.
- Netflix is able to create a large catalog of movies, "The Long Tail".



source: [Ra11]

It is not uncommon for sites to combine "free" content with "premium" (i.e., subscriber- or member-only) content, this is called ***freemium***.

"Give your service away for free, possibly ad supported but maybe not, acquire a lot of customers very efficiently through word of mouth, referral networks, organic search marketing, etc., then offer premium priced value added services or an enhanced version of your service to your customer base."

http://avc.com/2006/03/my_favorite_bus/

- The utility or "on-demand" model is based on **metering usage**, or a "**pay as you go**" approach.
- Unlike subscriber services, metered services are **based on actual usage rates**.
- Traditionally, metering has been used for essential services (e.g., electricity water, long-distance telephone services).
- Internet service providers (ISPs) in some parts of the world operate as utilities, charging customers for connection minutes, as opposed to the subscriber model common in the U.S.

Metered Usage

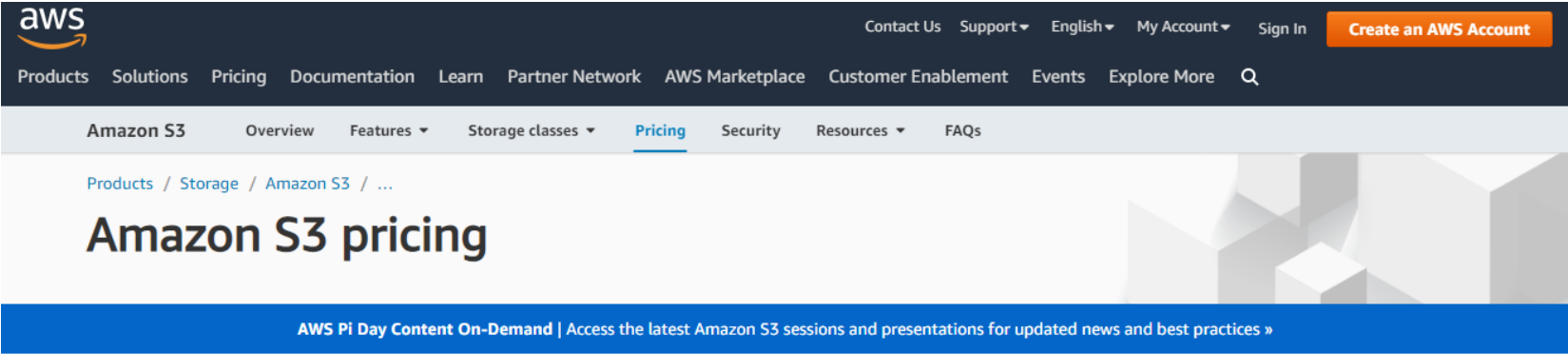
- Measures and bills users based on actual usage of a service (e.g., web hosting services).

Metered Subscriptions

- Allows subscribers to purchase access to content in metered portions (e.g., numbers of pages viewed without ads) (e.g., Slashdot).

source: [Ra11]

Amazon Web Services – S3 (File storage) - Pricing



Pay only for what you use. There is no minimum charge. There are six Amazon S3 cost components to consider when storing and managing your data—storage pricing, request and data retrieval pricing, data transfer and transfer acceleration pricing, data management and analytics pricing, replication pricing, and the price to process your data with S3 Object Lambda.

Storage

Requests & data retrievals

Data transfer

Management & analytics

Replication

S3 Object Lambda

You pay for storing objects in your S3 buckets. The rate you're charged depends on your objects' size, how long you stored the objects during the month, and the storage class—S3 Standard, S3 Intelligent-Tiering, S3 Standard-Infrequent Access, S3 One Zone-Infrequent Access, S3 Glacier Instant Retrieval, S3 Glacier Flexible Retrieval (Formerly S3 Glacier), and S3 Glacier Deep Archive. You pay a monthly monitoring and automation charge per object stored in the S3 Intelligent-Tiering storage class to monitor access patterns and move objects between access tiers. In S3 Intelligent-Tiering there are no retrieval charges, and no additional tiering charges apply when objects are moved between access tiers.

There are per-request ingest charges when using PUT, COPY, or lifecycle rules to move data into any S3 storage class. Consider the ingest or transition cost before moving objects into any storage class. Estimate your costs using the [AWS Pricing Calculator](#). To find the best S3 storage class for your workload, learn more [here](#).

Region:

Europe (Frankfurt)

	Storage pricing
S3 Standard - General purpose storage for any type of data, typically used for frequently accessed data	
First 50 TB / Month	\$0.0245 per GB
Next 450 TB / Month	\$0.0235 per GB
Over 500 TB / Month	\$0.0225 per GB

Community Model

- The viability of the community model is based on **user loyalty**.
- Users have a **high investment in both time and emotion**.
- Revenue can be based on the sale of ancillary products and services or voluntary contributions.

source: [Ra11]

Variations of the Community Model

Open Source (e.g., Red Hat, SUSE)

- Software developed voluntarily by a global community of programmers who share code openly.
- Instead of licensing code for a fee, open source relies on **revenue generated from related services like systems integration, product support, tutorials and user documentation.**

Social Networking Services (e.g., Facebook, Flickr)

- Sites that provide individuals with the ability to connect to other individuals along a defined common interest (professional, hobby, romance).
- Social networking services can provide opportunities for contextual advertising and subscriptions for premium services.

Knowledge Networks (e.g., Stackoverflow, Stack Exchange)

- Discussion sites that provide a source of information based on the sharing of expertise among professionals.

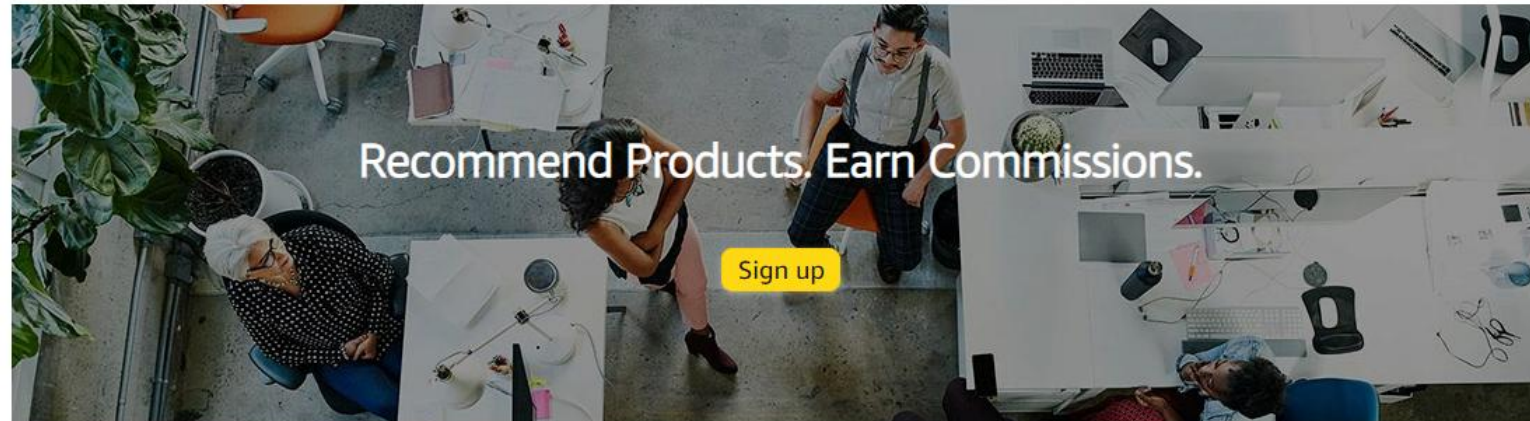
Open Content (e.g., Wikipedia)

- Openly accessible content developed collaboratively by a global community of voluntary contributors.

source: [Ra11]

- In contrast to the generalized portal, which seeks to drive a high volume of traffic to one site, the affiliate model provides **purchase opportunities wherever people may be surfing**.
- It does this by offering **financial incentives** (in the form of a revenue percentage) to affiliated partner sites.
- The affiliates provide **purchase-point click-through** to the merchant.
- It is a **pay-for-performance** model.
- If an affiliate does not generate sales, it represents no cost to the merchant.
- The affiliate model is inherently well-suited to the web, which explains its popularity.
- **Variations:**
 - Pay-per-click: Site that pays affiliates for a user click-through.
 - Revenue Sharing: offers a percent-of-sale commission based on a user click-through in which the user subsequently purchases a product.
- Examples: Amazon.com, Barnes & Noble

source: [Ra11]



Amazon Associates - Amazon's affiliate marketing program

Welcome to one of the largest affiliate marketing programs in the world. The Amazon Associates Program helps content creators, publishers and bloggers monetize their traffic. With millions of products and programs available on Amazon, associates use easy link-building tools to direct their audience to their recommendations, and earn from qualifying purchases and programs.



1

Sign up

Join tens of thousands of creators, publishers and bloggers who are earning with the Amazon Associates Program.



2

Recommend

Share millions of products with your audience. We have customized linking tools for large publishers, individual bloggers and social media influencers.



3

Earn

Earn up to 10% in associate commissions from qualifying purchases and programs. Our competitive conversion rates help maximize earnings.

Data about consumers and their buying habits are extremely valuable.

Especially when that information is carefully analyzed and used to **target marketing campaigns**. Some firms are able to function as infomediaries by collecting and selling information to other businesses.

Audience Measuring Services (e.g., Nielsen Online)

- Online audience market research agencies

Incentive Marketing (e.g., PAYBACK)

- Customer loyalty program that provides incentives to customers such as redeemable points or coupons for making purchases from associated retailers.
- Data collected about users is sold for targeted advertising.

Advertising Networks (e.g., PropellerAds)

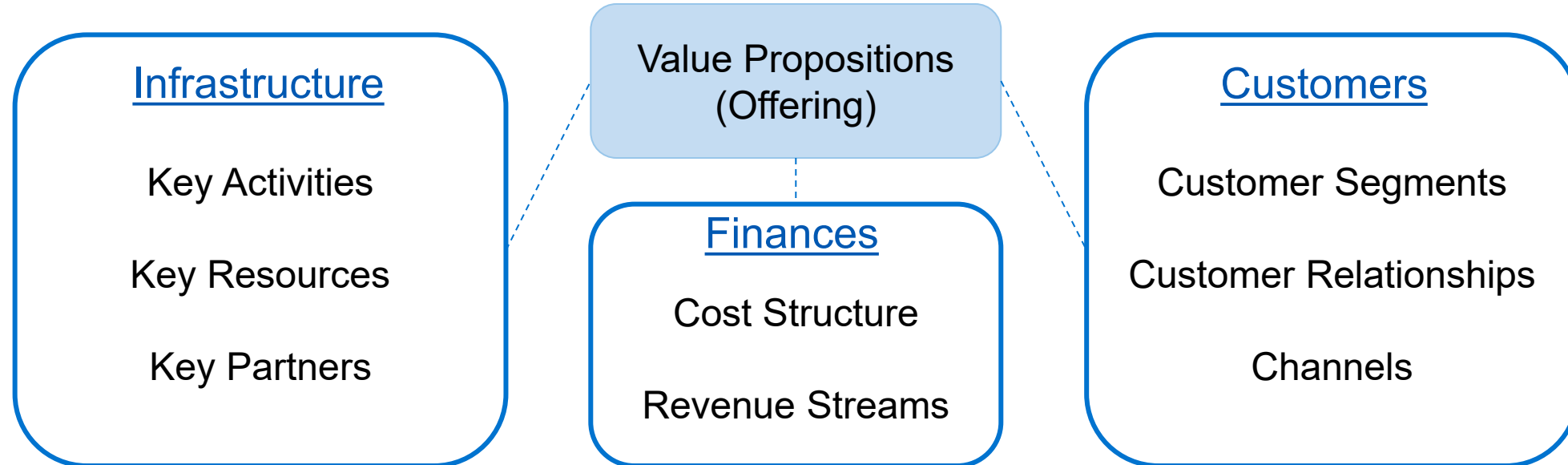
- Feed banner ads to a network of member sites, thereby enabling advertisers to deploy large marketing campaigns
- Ad networks collect data about web users that can be used to analyze marketing effectiveness

1. Web Site Genres

- Business Models on the Web
- Business Model Canvas & Value Proposition Canvas

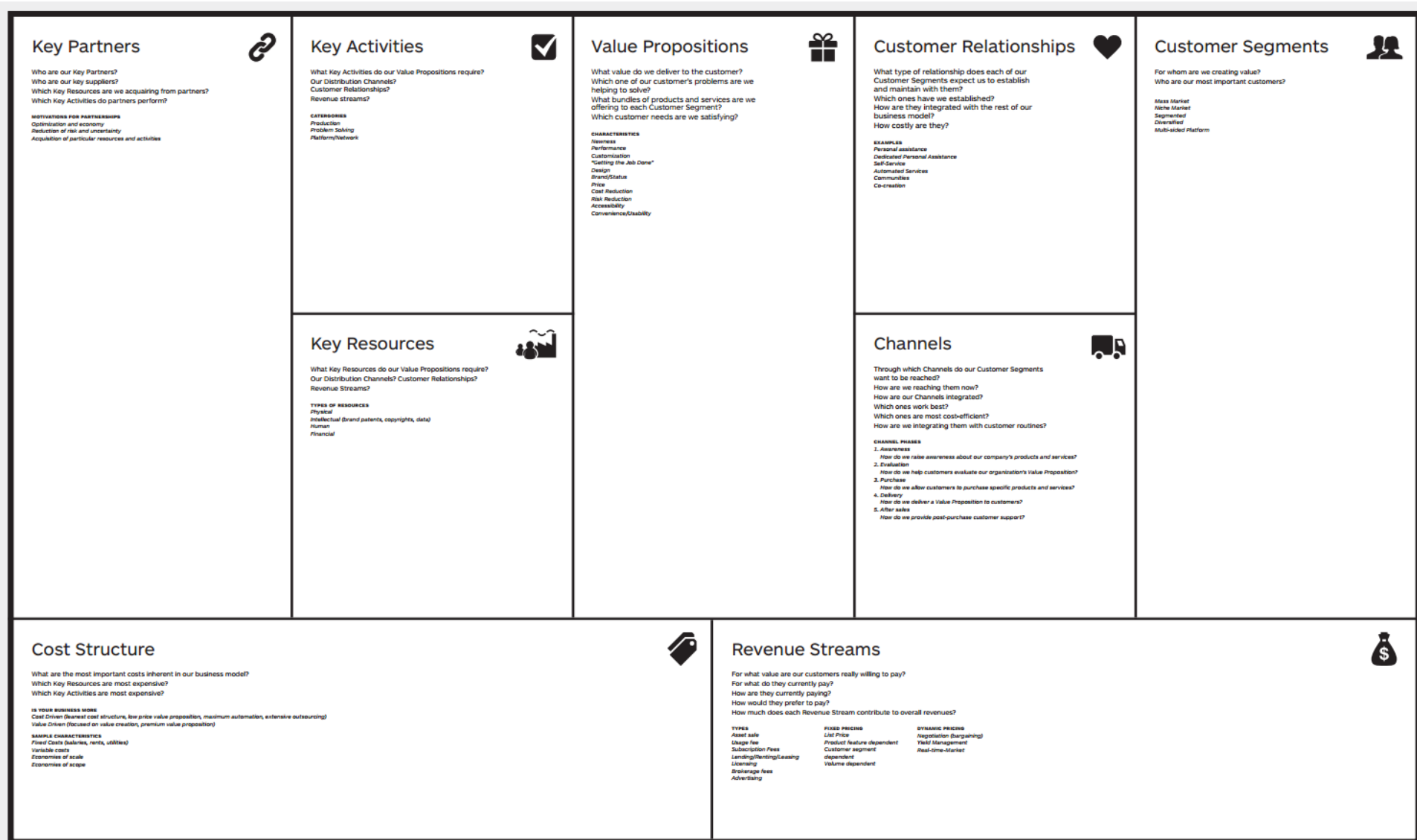
Business Model Canvas (BMC) & Value Proposition Canvas (VPC)

- Strategic management tool used to develop, document and evaluate business models.
- Why? Force clear definition / Present to others / Identify problems early / Stay on track
- Developed by Alexander Osterwalder and Yves Pigneur and initially presented in their book "Business Model Generation".
- Nine building blocks that focus on separate segments of a business model:



- VPC focuses on the two blocks "Customer Segments" and "Value Propositions".

Business Model Canvas (BMC)



Watch the video for more information [<https://www.strategyzer.com/canvas/business-model-canvas>]

Customer Segments

For whom are we creating value?
Who are our most important customers?

Mass Market
Niche Market
Segmented
Diversified
Multi-sided Platform

Value Propositions

What value do we deliver to the customer?
Which one of our customer's problems are we helping to solve?
What bundles of products and services are we offering to each Customer Segment?
Which customer needs are we satisfying?

CHARACTERISTICS

Newness
Performance
Customization
"Getting the Job Done"
Design
Brand/Status
Price
Cost Reduction
Risk Reduction
Accessibility
Convenience/Usability



Watch the video for more information [<https://www.strategyzer.com/canvas/business-model-canvas>]

Customer Relationships Channels

What type of relationship does each of our Customer Segments expect us to establish and maintain with them?
Which ones have we established?
How are they integrated with the rest of our business model?
How costly are they?

EXAMPLES

Personal assistance
Dedicated Personal Assistance
Self-Service
Automated Services
Communities
Co-creation

Through which Channels do our Customer Segments want to be reached?
How are we reaching them now?
How are our Channels integrated?
Which ones work best?
Which ones are most cost-efficient?
How are we integrating them with customer routines?

CHANNEL PHASES:

1. *Awareness*
How do we raise awareness about our company's products and services?
2. *Evaluation*
How do we help customers evaluate our organization's Value Proposition?
3. *Purchase*
How do we allow customers to purchase specific products and services?
4. *Delivery*
How do we deliver a Value Proposition to customers?
5. *After sales*
How do we provide post-purchase customer support?

Watch the video for more information [<https://www.strategyzer.com/canvas/business-model-canvas>]

Key Partners

Who are our Key Partners?
Who are our key suppliers?
Which Key Resources are we acquiring from partners?
Which Key Activities do partners perform?

MOTIVATIONS FOR PARTNERSHIPS:

Optimization and economy
Reduction of risk and uncertainty
Acquisition of particular resources and activities

Key Activities

What Key Activities do our Value Propositions require?
Our Distribution Channels?
Customer Relationships?
Revenue streams?

CATEGORIES

Production
Problem Solving
Platform/Network

Key Resources

What Key Resources do our Value Propositions require?
Our Distribution Channels? Customer Relationships?
Revenue Streams?

TYPES OF RESOURCES

Physical
Intellectual (brand patents, copyrights, data)
Human
Financial

Watch the video for more information [<https://www.strategyzer.com/canvas/business-model-canvas>]

Cost Structure

What are the most important costs inherent in our business model?
Which Key Resources are most expensive?
Which Key Activities are most expensive?

IS YOUR BUSINESS MORE:

Cost Driven (leanest cost structure, low price value proposition, maximum automation, extensive outsourcing)
Value Driven (focused on value creation, premium value proposition)

SAMPLE CHARACTERISTICS:

Fixed Costs (salaries, rents, utilities)
Variable costs
Economies of scale
Economies of scope

Revenue Streams

For what value are our customers really willing to pay?
For what do they currently pay?
How are they currently paying?
How would they prefer to pay?
How much does each Revenue Stream contribute to overall revenues?

TYPES:

Asset sale
Usage fee
Subscription Fees
Lending/Renting/Leasing
Licensing
Brokerage fees
Advertising

FIXED PRICING

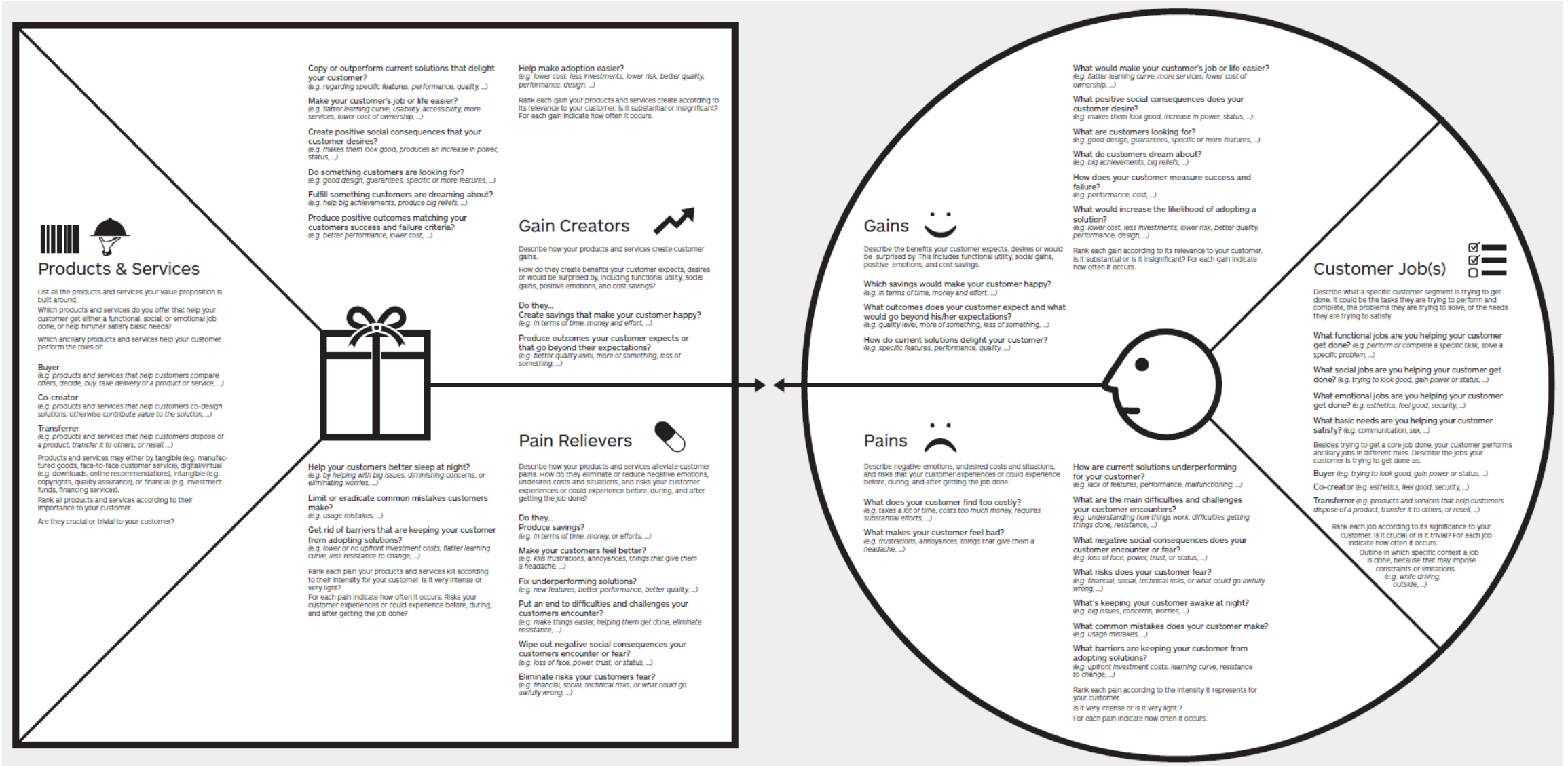
List Price
Product feature dependent
Customer segment dependent
Volume dependent

DYNAMIC PRICING

Negotiation(bargaining)
Yield Management
Real-time-Market

Watch the video for more information [<https://www.strategyzer.com/canvas/business-model-canvas>]

Value Proposition Canvas (VPC)



[<https://www.strategyzer.com/canvas/value-proposition-canvas>]